

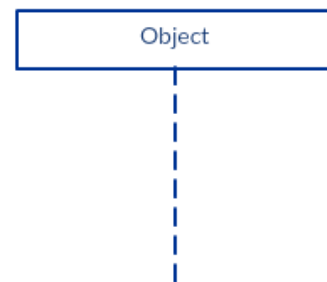
Sequence Diagrams: - A Quick Overview

The Actor, Object and Lifelines

A sequence diagram is made up of several symbols. The **actor** symbol and name is placed above the **lifeline** which is shown as a **perforated vertical line**. A lifeline notation is used with an actor symbol when the sequence diagram is owned by a use case.



There can be a number of **objects** or **parts that interact with each other** or the actor during the sequence. The actor and object elements should be arranged horizontally across the top of the diagram. Each element should be named. The notation for the object should be a colon (:) followed immediately by the object name. This combination should be underlined e.g. **:ATM system**



No two lifelines should overlap each other.

Message Arrows

A **message** is indicated by means of an **arrow** which can flow from an actor to object or object to object(s). A message can flow in any direction; from left to right or right to left. You can describe the message being sent from one object to the other on the arrow by **using different arrowheads** to indicate the type of message being sent or received.

- *A Synchronous message*

A synchronous message uses the solid arrowhead shown below and it indicates that **the sender waits for the receiver to process the message** and return before carrying on with another message.

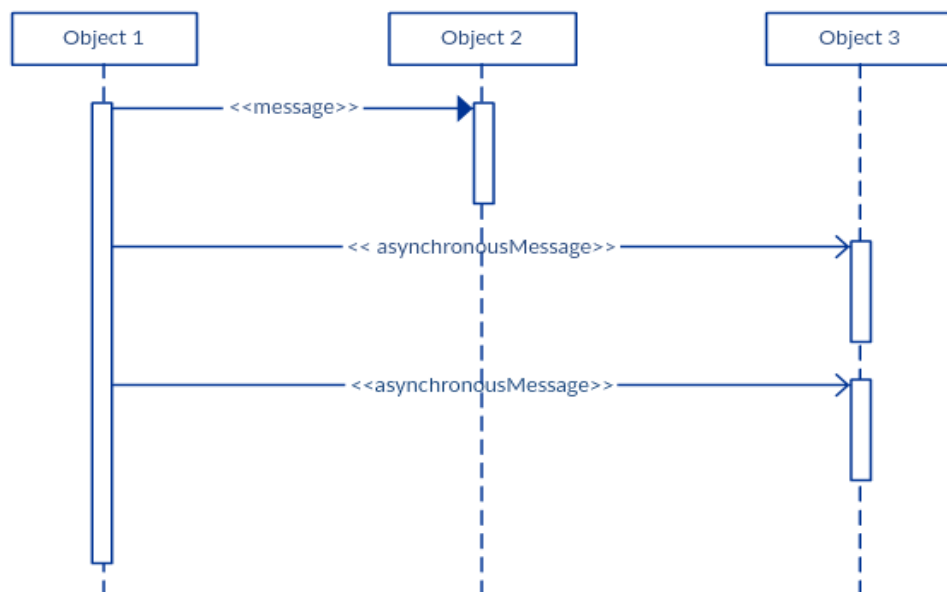


An Asynchronous message

An asynchronous message is used when **the message caller does not wait** for the receiver to

process the message and return before sending other messages to other objects within the system. A

line arrowhead is used to show this type of message.

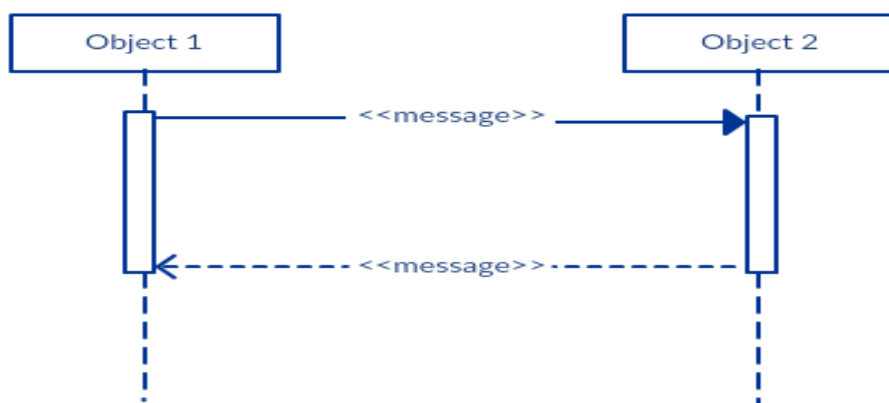


The message arrow comes with a description on it known as a **message signature**. The format for this message signature is presented below. **The message_name** is **mandatory** and the other clauses are optional.

attribute = message_name
(arguments): return_type

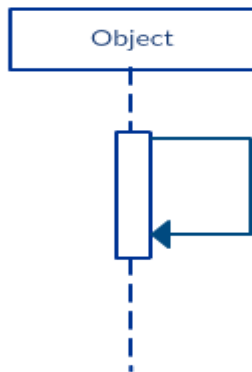
- *Return message*

A **return message** is used to indicate that the message receiver is done processing the message and is **returning control** over to the message caller. The return message is indicated by a perforated line and a line arrowhead.



- **Reflexive message**

A **reflexive message** occurs when an **object sends a message to itself**. It is indicated with a message arrow that starts and ends at the same lifeline as shown in the example below.



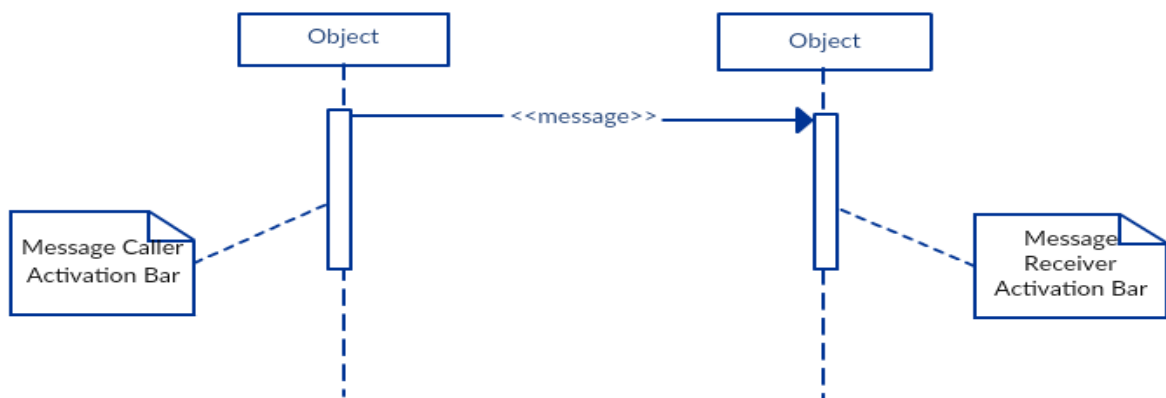
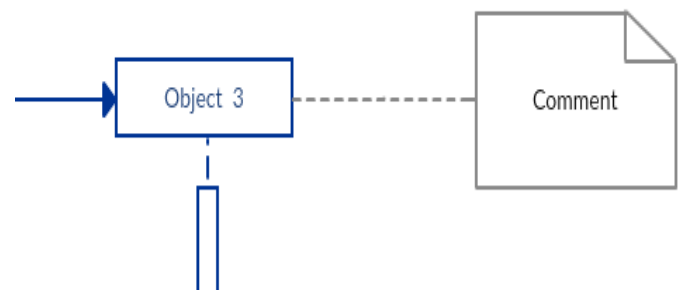
sequence diagram, an interaction between two objects occurs when one object sends a message to another. The use of the activation bar on the lifelines of the Message Caller (the **object that sends** the message) and the Message Receiver (the **object that receives** the message) indicates that **both are active or instantiated** during the exchange of the message.

Comment

The **comment** object is a rectangle with a **folded-over corner** and it can be linked to the related object with a perforated line or an activation bar.

Activation Bars

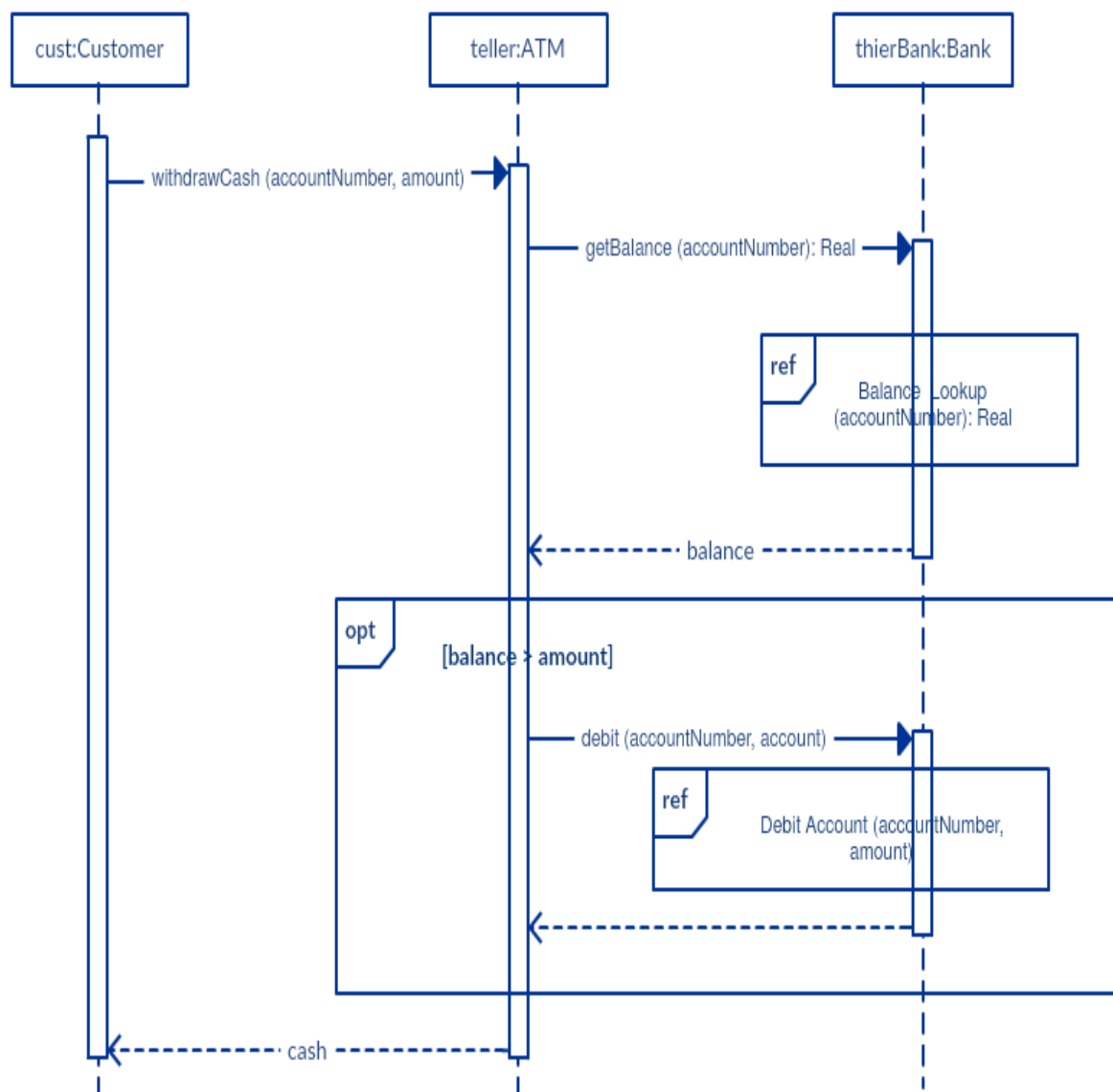
Activation bar is the box placed **on the lifeline**. It is used to indicate that an object is **active (or instantiated)** during an interaction between two objects. The length of the rectangle indicates the duration of the objects staying active. In a



Loops

Loop fragment is used to represent a **repetitive sequence**. Place **the words 'loop'** in the **name box** and **the guard condition** near the **top left corner of the frame**.

The guard condition in a **loop fragment** can be a Boolean test, or it can test two other special conditions. These are the **minimum iterations** (written as *minint* = [the number]) and **maximum iterations** (written as *maxint* = [the number]). If it is a minimum iterations guard, the loop must execute **not less than the number mentioned**, and if it is a maximum iterations guard, the loop **mustn't execute more than** the number indicated.



Adapted from <https://creately.com/blog/diagrams/sequence-diagram-tutorial/>